

Demand Sensitive Bus Dispatching System

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The phenomenon of bus bunching occurs all over the world and is a large problem for transportation service providers. It occurs when two buses which are scheduled to have a headway in between them end up bunching, moving together along the route. It creates an inconvenience to the passengers and causes increased waiting times and travel times. This research will investigate the use of Reinforcement Learning as a solution to this problem and compare it with a Rule Based control strategy. This will be tested by simulating a bus corridor using traffic simulation software. The actions will consist of a combination of bus holding, stop skipping and boarding limits. We will compare the different approaches by evaluating the on-time performance and the average headway for each approach.

1 INTRODUCTION

The effectiveness and reliability of a public transportation system is key to have a more sustainable mode of frequently used urban transportation for passengers. Thus, having a system that works according to the demand is not only critical for the passengers but also to the traffic in general. These types of public transportation services are subject to many disturbances given the stochastic nature of the environment in which they operate. Such disturbances are caused by traffic congestion, traffic accidents as well as vehicle capacity, route lengths and most importantly variability in passenger demand. These all have a negative impact on waiting times - which is the time spent by the passengers waiting for their bus to arrive, dwell times - which is the time taken for passengers to board and alight the bus and travel times - which is the time a passenger spends travelling since boarding the bus to alighting the bus at their destination (this includes the dwell time taken at each intermediate stop).

Delay is likely to worsen along the route. A delayed bus is expected to be met by a larger number of passengers waiting at future stops therefore increasing the dwell time at each future stop. On the other hand, the next bus will then have less passengers to pick up and will hence have significantly less dwell and travel time as it almost catches up with the previously delayed bus. This will create headway instability (or variability) between the two buses, also known as bus bunching which is the phenomenon of two buses who are scheduled to have a headway of a predetermined amount of minutes but end up bunching together along the route. This phenomenon is also referred to by other works as the Bangkok Effect [1], Bus Platooning [2] or Vehicle Pairing. [3][4]. Figure 1 shows two scenarios, where on top, is the ideal scenario with little to none headway variability and on the bottom, is the bunching bunching.

There are two approaches to the bus bunching problem in similar studies. The first is that bus bunching is an effect of a bad transport infrastructure (which are often alleviated by introducing signal priority and bus lanes) and the second is that bus bunching is the main problem that must be solved (which is often alleviated by adjusting schedules or giving live control actions) [4]. This study will be focusing on the latter.

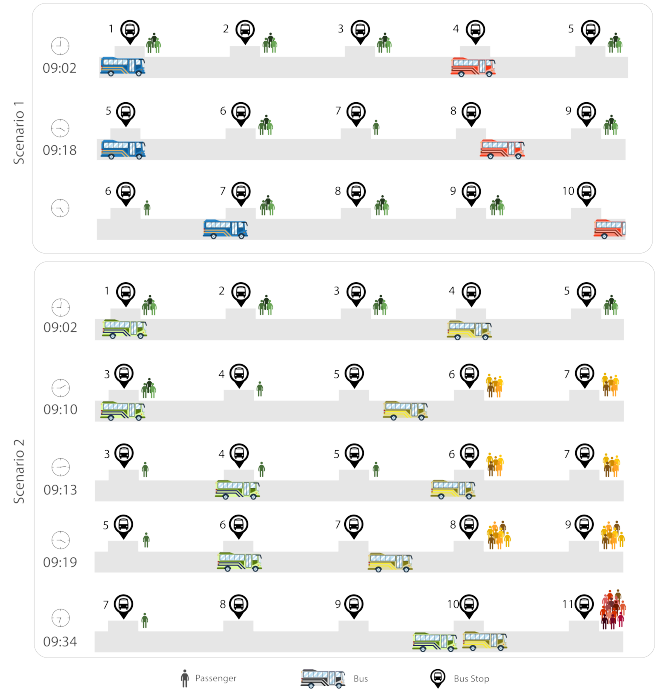


Fig. 1. Bus Bunching Illustration (Source: [4])

Scenario 1: Ideal Scenario, Scenario 2: Bus Bunching

2 AIMS AND OBJECTIVES

The aim of this study is to investigate whether an A.I. algorithm can help alleviate the problem of the bus bunching phenomenon. The objectives of this research are to:

- (1) Simulate the bus bunching phenomenon using traffic simulation software (No Control)
- (2) Implement Rule-Based Strategies on the same simulation and measure their effect (Rule-Based Control)
- (3) Apply Reinforcement Learning (RL) Techniques in order to dynamically control the bus system and compare it to the other two control strategies.

The above mentioned control strategies will be compared with each other using the performance metrics found in literature detailed in Section 3.4.

3 BACKGROUND AND LITERATURE REVIEW

Headway variability is the main way in which bus service regularity is calculated. It is defined by;

$$var(H) = \frac{1}{N} \cdot \sum_{i=1}^N (AH_i - SH_i)^2 \quad (1)$$

where AH = actual headway and SH = scheduled headway. [5]

3.1 Variables that affect Headway Variability

Tirachini et al [6] compared multiple studies in order to find the most influential variables with respect to headway variability. A flawed design of the route could be a contributor to the bus bunching problem. For example if it is too long or the travel speed is too low, the scheduled frequency might not keep up with the average demand. If the scheduled frequency is too high, then it is more likely that buses will end up bunching together [7]. Consequently, an ideal bus dispatching strategy must be found such that the phenomenon is avoided and at the same time, passengers do not experience long waiting times. The optimal strategy takes into account the demand variability and traffic patterns.

Other variables that are of influence to the headway variability of a route include the distance travelled since the beginning of the route. Naturally, due to the unpredictability of accidents and road irregularities that the vehicle encounters, there is an increased likelihood of the vehicle bunching with its leading vehicle as it moves along the route [8]. Traffic congestion increases travel time significantly and as a result, the delay of the bus increases [9].

Passenger demand is another contributor that has an impact on headway variability. The number of passengers that board and alight a vehicle is proportional to the dwell time of that vehicle. A possible aid to this issue would be to introduce off-board payment machines so that the alighting process is sped-up [10]. Boarding and alighting rules could also be implemented in order to reduce the friction between the two types of passengers, and hence reduce the dwell time. Despite the effects these measures have on reducing dwell time, it can never be diminished as it is proportional to the demand.

3.2 Negative Effects of Headway Variability

There are many negative effects that emerge from headway variability which ultimately lead to the phenomenon of bus bunching. Let w represent the actual waiting time of a passenger where $0 < w < h$ and h is the headway between the last bus and the next. Owsuna and Newell [11] define the expected waiting time $E(w)$ of a passenger at a stop as

$$E(w) = \frac{E(h)}{2} \left(1 + \left(\frac{\sigma(h)}{E(h)} \right)^2 \right) \quad (2)$$

where; $\sigma(h)$ represents the standard deviation of headways at the stop. This would mean that, in an ideal situation with equal headways, i.e. $\sigma(h) = 0$, and randomly distributed passenger arrivals, the average waiting time of a passenger is half the headway.

Hollander and Liu [12] found that passengers would much rather have less waiting time on bus stops over less in-bus travel time. Poor quality of service has the potential to persuade users to resort to other modes of transportation that are more reliable. Bunching therefore creates a bad reputation to the rest of society since dissatisfied passengers will start advocating against the use of public transport and its lack of quality service. The findings in [13] support this. A survey of Maltese bus users before and after a bus service reform and found that many of the respondents who were car users seemed to have a more negative attitude towards bus use than actual bus users [13]. When analysing the reasons for such attitude one of the characteristics which were deduced to be of concern to the bus users was in fact time [13]. This negative reputation led to more

bus users shifting to other modes of transportation such as cars or taxis as which translated to a high increase in car users.

3.3 Control Strategies

Various control strategies have been proposed in order to reduce the problem of bus bunching.

3.3.1 Stop Skipping. This strategy allows for some vehicles to only visit a fraction of the stops along a route, thus reducing the travel times as well as dwell times. Zonal Express Service is an operational strategy in which a bus will visit all the stops in a particular zone and then skip all stops until reaching the next zone [14]. This is useful when applied to routes that have segments which have high demand as well as segments which have low demand. The stop-skipping strategy increases the waiting time of the passengers who are waiting at bus stops which are skipped. On the other hand, this strategy is favourable to the passengers who boarded the bus at the beginning of the route and intend on alighting at the end and restores the headway to the next bus [15].

3.3.2 Bus Holding. This strategy aims at regulating the departure time of vehicles from stops. Time points (TPs) are the stops at which holding may take place and typically a bus route will only have a few of them [16]. A bus is likely to be held if it is ahead of schedule or its headway with respect to the bus before it is too short. The obvious advantage of bus holding is that headway can be easily manipulated. Also, since the bus will be held at a bus stop, all the passengers who arrive while the bus is being held will have no waiting time. The passengers who are already in the bus will still suffer from increased in-vehicle times as well as increased travel times. There are two types of holding methods that fall under Naive Methods [16]; the schedule-based method holds a bus until its departure time is reached and the headway-based method holds a bus until the forward or backward headway reaches a predetermined threshold [17][18]. They are referred to as Naive methods because they are simple to carry out. The former of the two methods is calculated by subtracting the vehicle's actual arrival time from scheduled arrival time while the latter is calculated by obtaining the departure time of the previous bus from the current control point. If it is less than a predetermined threshold, i.e. there is a short headway, the bus will be held. The other two types of holding methods are partial holding methods and prediction based holding methods. Xuan et al [19] implemented a partial holding method by setting the forward headways according to the scheduled headway and tweaking it using a factor β and according to the off-schedule time (arrival time subtracted by scheduled departure time) tweaked using a factor α where $0 < \alpha < 1$. Similar to Naive Methods, this method requires knowing the scheduled headway and the scheduled departure time. An advantage of this method over naive holding methods is that the suggested holding time is only a fraction of the threshold. Prediction-based holding methods work by holding a bus according to the predicted arrival time of the following bus. This method requires real-time data from Automatic Vehicle Locators (AVLs) in order to make such predictions, and such data is becoming increasingly common in many public transportation systems. Real-time data was

used to produce sophisticated models that are able to predict arrival times accurately and efficiently [20][21][22].

3.3.3 Boarding Limit. This strategy constrains the number of passengers that are allowed to board the vehicle at a given stop [23]. Such limits are applied even when the vehicle is not at full occupancy. This strategy has similar effects to stop-skipping as it aims at improving the operational speed and efficiency, decreasing the dwell time. This strategy is already sometimes implemented by bus drivers when they notice that the bus following them is close to their tail. In reaction to bus bunching, the drivers instruct boarding passengers to wait for the next bus. Many passengers often prefer waiting to board a slightly emptier bus as long as they know that their waiting time will be brief [23].

3.4 Existing A.I. Techniques

In Reinforcement Learning (RL), the agent learns by interacting with the environment. RL is based on the reward hypothesis; which states that all goals can be described by the maximisation of the expected cumulative reward. At each time step t it takes as input some representation of the environment state S_t and according to some learned policy, π , an action a is selected from the set of all possible actions $A(S_t)$ under that state. The value function of a state s under a policy π , denoted by $V_\pi(s)$ is the expected reward when starting from state s and following π thereafter. More formally, the *state-value function* v_π for policy π can be defined as;

$$V_\pi(s) = Q_\pi(s, \pi(s)) \quad (3)$$

The value of taking action a in state s under a policy π , referred to as the *action-value function*, denoted by $q_\pi(s, a)$, is defined as;

$$Q_\pi(s, a) = R(s, a) + \gamma \cdot \sum_{s' \in \text{succ}(s, a)} T(s, a, s') \cdot V_\pi(s') \quad (4)$$

where; γ is the discount rate used in order to determine how much weight the agent should give to a future reward over an immediate reward and $R(s, a)$ is the immediate action reward.

Bus holding control is a very common and effective strategy used in literature implementing Reinforcement learning as their solution to bus bunching. The authors in [24] apply multiple Q-Learning agents using bus-holding and compare its advantages with traditional rule-based holding control. The applied the agents on a series of signalised intersections and reduced bunching by 88%. Wang and Sun [25] developed an asynchronous multi-agent reinforcement learning model (ASMR) in order to reduce bus bunching by holding control. The authors in [25] propose a credit assignment framework for asynchronous control (CAAC) by integrating actions from other agents and use a neural network to carry out inductive learning for policy evaluation. They compared their results to traditional rule based holding control methods as well as existing multi-agent reinforcement learning (MARTL) methods and their proposed model outperformed both.

We will investigate Deep Reinforcement Learning Algorithms as a solution to this problem. Such algorithms include Deep Deterministic Policy Gradients (DDPG) which is an actor-critic algorithm designed for use in environments with continuous action spaces. This algorithm learns a Q-function and a policy simultaneously and uses the off-policy data along with the bellman equation to learn

the Q-function [25]. It in turn uses that function to learn the policy. Another algorithm worth investigating is the Twin Delayed DDPG (TD3) which is an improvement over the DDPG as it learns two Q-functions and uses the smaller Q-value to create targets in the loss function. It is referred to as an improvement because DDPG can sometimes overestimate Q-values, leading to a bad policy [26].

3.5 Performance Metrics

In the works covering the bus bunching problem, there is no general consensus on how to measure the headway variability and what metric should be used to analyse the performance of the proposed model.

- *Measures expressed with respect to time* are simple to implement but can be deceitful when comparing routes or segments of routes that have different characteristics. These measures include comparing the actual arrival/departure time of a bus with the scheduled arrival/departure time. Other time related measures can be with respect to waiting time as well as travel time.
- *Percentage Measures.* These measures are the percentage of observations that fall within a predetermined range such as the scheduled time or the expected time. For example: showing that 30% of the actual headways are more than the scheduled headway. Or that 40% of busses' travel times are almost double the scheduled travel time. The authors in [10] found that this method is not very accurate as it operates using an 'all-or-nothing assessment' of a particular observation in order to determine if it is acceptable or not. A possible remedy to this drawback, which is to have multiple ranges under which an observation can be classified. In this way, one can gain a greater idea of the performance [10].
- *Variability Measures.* These can include
 - *Variance of Headways*,
 - *Standard Deviation of Headways*, and
 - *Headway Coefficient of Variation CV(h)* which is defined as the ratio between the standard deviation and the mean actual headway. An irregular service will result in a high CV(h). Despite this metric giving a clear idea of the headway variation, it does not give a clear idea of the passenger's experience with respect to waiting time [27][28]. This is especially useful when compared with the first two variability measures since it is a ratio, therefore it can be used to compare two very different headways [10].
- *Reliability Measures* are those which are concerned with the feelings of the passengers and their perspective. The most common are:
 - *On-Time Performance (OTP).* This is widely used in transportation research [29][27]. It is also applied in real life situations [30][31][32]. A service is said to be on time if it is between d minutes early and e minutes late [6]. Many transportation services define d to be 1 minute and e to be between 4 to 5 minutes [30][33]. This metric can only be applied to services that operate using a fixed schedule, versus services that are scheduled to operate frequently, and works best when used to evaluate services that are

scheduled to have long headways. This is because a passenger will often notice that a vehicle is not on-time after having waited for at least 10 minutes [10].

- *Excess wait time (EWT)* is a metric that is defined by the relationship between a passengers' arrival time at a bus stop, a bus' scheduled departure time and its actual departure time. This is calculated by subtracting the scheduled departure time from the actual departure time, thus showing the extra wait time a passenger would endure.

3.6 Traffic Simulators

In order to be able to represent a real-world environment accurately, a good traffic simulator is required. Such a simulator must be able to provide accurate results from which meaningful conclusions can be made. It is important that such a simulation environment can emulate real life as much as possible by allowing the user to alter the demand and add in traffic freely. The three classifications of traffic simulation models are microscopic models, macroscopic models and mesoscopic models. Macroscopic models do not allow for details of individual vehicles to be different as they consider traffic as a whole (taking into consideration its fluidity, density etc..) whereas microscopic models allow for each vehicle to have its own attributes and give the user full control over its behaviours [34]. Their use is mainly aimed at collecting data such as the speed, the travel time and headway of a vehicle. Such simulators therefore allow the user to implement their own scheduling and routing algorithms based on these values [35]. Mesoscopic models are a combination of microscopic and macroscopic. They allow for the behaviour of individual vehicles as well as the behaviour of traffic in general to be manipulated [36]. Nowadays several Traffic Simulators exist, with the main three being used in literature being Simulation of Urban Mobility (SUMO)¹, Multi-Agent Transport Simulator (MATSim) [37] [38] [39] and the Flexible Mobility Services Testbed developed by [40].

4 PROPOSED METHODOLOGY

We intend to create a case study that simulates the phenomenon and the proposed control strategies in an imaginary bus corridor like in the work done by [41]. SUMO was chosen for this project for multiple reasons:

- (1) It is open-source
- (2) It provides on-line interaction with the simulation via Traffic Control Interface (TraCI) API
- (3) It provides full microscopic control over the vehicles and network.
- (4) It is updated regularly whereas the Flexible Mobility Testbed was last updated in 2014 and has only been used by [40].
- (5) It allows for vehicle behaviour changes mid-route whereas MATSim only allows interruption when the vehicle is stopped.

The application does not require any programming but rather takes XML files as input for the environment and route variables. In order to be able to interact with the simulation, TraCI allows for multiple programming languages to be used. For this project, we will be making use of Python. And any and all statistical data that is

outputted will be handled through the use of CSV files. SUMO will be used to create the network and all its environmental variables along with TraCI in order to be able to manipulate and retrieve data from the simulation using python. The network will consist of 24 evenly spaced bus stops and the route will contain 14 buses having a maximum passenger capacity of 70 and travelling at 17km/hr. The passenger flow will follow the distribution as in [41] (Figure 2).

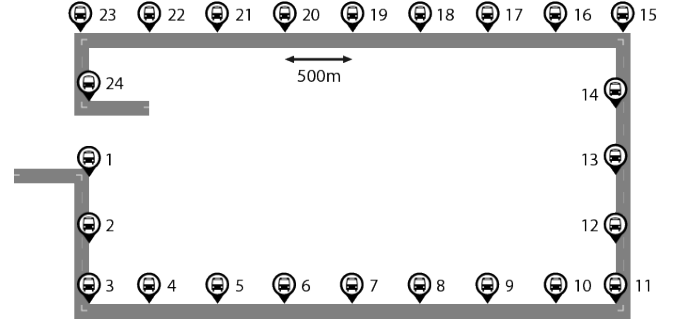


Fig. 2. Bus Corridor as implemented in [41].

(1) Simulating the problem (No Control)

We intend to run the simulation once without control, and observe bus bunching take place during the normal running of the network, especially around peak-demand bus stops i.e. bus stops 10-16.

(2) Implementing Rule Based Control

Then a set of rules will be formulated and applied when satisfying certain criteria. For Example: when the forward headway is less than 3 minutes, hold the bus for 3 minutes. This will be referred to as Rule Based Control.

(3) RL Based Control

Finally AI Techniques, more specifically RL techniques, will be applied in order to implement Dynamic Control. In order to carry this out, the simulation will be run numerous times and during each simulation step the RL will receive preprocessed data and come up with policies according to that data. It will then use performance metrics as rewards and improve its policies according to the reward. The model will keep training until the average performance over the last ten episodes is very similar to the average performance of the last 50 as well as the last 100 episodes, i.e. it converges. We hypothesise that this will be the best control method out of all three.

5 EVALUATION

We intend to evaluate the effectiveness of the control strategies using a number of variables. These include the average fleet occupancy level (%), the average passenger wait and travel times (minutes) and the average forward headway (minutes). We will also compare the on-time performance OTP using equation 5 where a vehicle will be considered on time if it is between 1 minute early and 4 minutes late. The EWT will be calculated using equation 6.

$$OTP = \frac{No.OfOnTimeVehicleArrivals}{TotalNumberOfVehicleArrivals} \cdot 100\% \quad (5)$$

¹<https://www.eclipse.org/sumo/>

$$EWT = AWT - SWT \quad (6)$$

where; EWT = excess wait time, AWT = actual wait time and SWT = scheduled wait time. [5]

$$AWT = \frac{\sum_{i=1}^n AH_i^2}{2 \cdot \sum_{i=1}^n AH_i}; SWT = \frac{\sum_{i=1}^n SH_i^2}{2 \cdot \sum_{i=1}^n AH_i} \quad (7)$$

We will use plot graphs displaying the variable outcomes for each control strategy using python's matplotlib library.

6 CONCLUSION

This report details the research which has been conducted so far as well as the preliminary work required to set up the simulation. The report outlined the concept of bus bunching and the causes behind it. It also reviewed similar works which have investigated the problem. The anticipated result of this Final Year Project is to provide a comparison of solutions to the bus bunching problem shown via a simulation, similar to the one created by Delgado in [41]. The implementation and experimentation is intended to follow the timeline depicted in figure 3 in Appendix A. The results of the experiments will be fairly evaluated and compared with two benchmark strategies as well as strategies implemented in similar works.

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A GANTT CHART

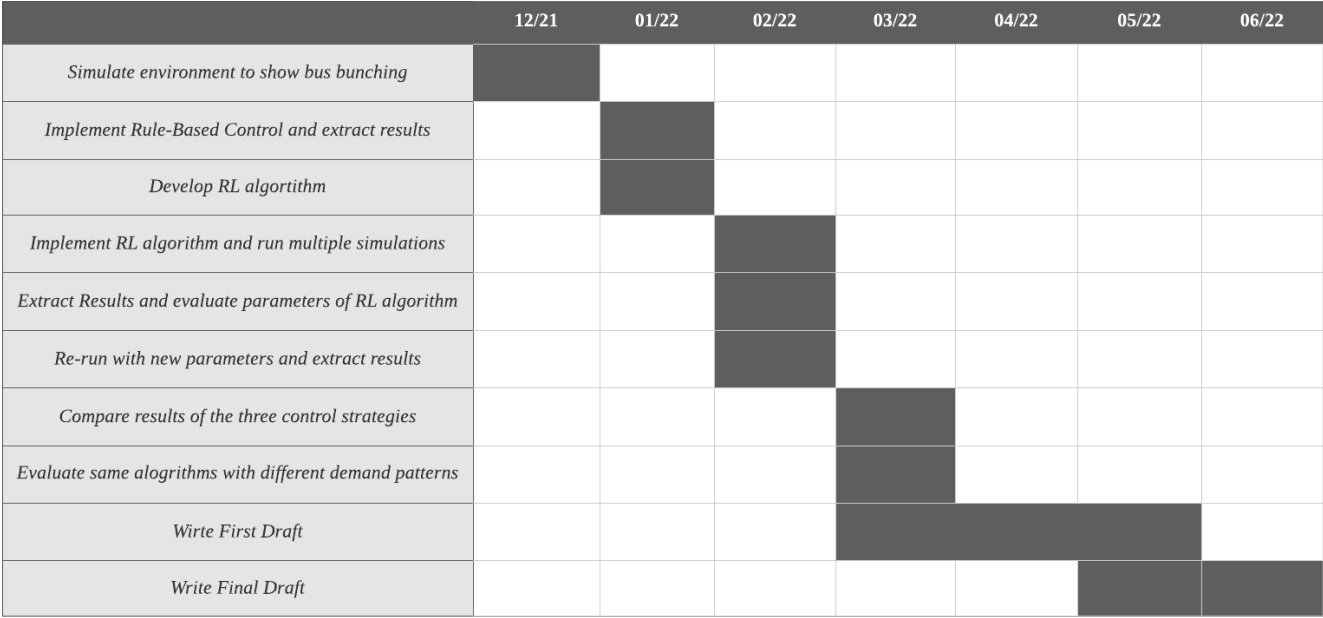


Fig. 3. Gantt Chart Presenting the Proposed Timeline